THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): It is easier to push a box on a floor with rollers than on a rough floor. Reason (R): Rolling friction is less than sliding friction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A box of mass 20 kg is pulled with a force of 60 N along a floor where friction is 40 N. The net force is: 1

(a) 100 N **(b)** 20 N **(c)** 60 N **(d)** 40 N

3. A tug-of-war rope is stationary. If team A suddenly pulls harder, then the rope: 1

(a) moves towards team A **(b)** stays at rest
(c) breaks always **(d)** moves towards team B

4. Assertion (A): A sky-diver falls at constant speed after opening the parachute. Reason (R): The forces on her are balanced. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A man weighs 600 N on Earth. On a planet where gravity is 1/3 of Earth's, his weight is: 1

(a) 200 N **(b)** 300 N **(c)** 600 N **(d)** 1800 N

6. A magnet is brought near an iron nail and the nail jumps to the magnet. The force that does the work is: 1

(a) friction **(b)** electrostatic force
(c) magnetic force **(d)** gravity

7. Assertion (A): A body of mass 5 kg has the same mass on the Moon. Reason (R): Mass depends on the gravitational pull at that place. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A 60 kg astronaut on the Moon has mass: 1

(a) 360 kg **(b)** 0 kg **(c)** 60 kg **(d)** 10 kg

9. **Assertion (A):** Friction is a contact force. 1
Reason (R): Friction acts only when two surfaces are in contact and opposes their relative motion.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The mass of an object is the same on the Earth and on the Moon. 1
Reason (R): Mass is the amount of matter in an object, which does not change with place.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A body of mass 3 kg has a weight of about 30 N on Earth. What is its mass and weight on the Moon if gravity there is one-sixth? 2

12. Why does a block slide down a slope faster when the slope is polished? 2

13. State why heavy trucks have more tyres. 2

14. Explain how a charged rod can attract a neutral object. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A box on a floor is pushed with 30 N and moves with a constant velocity. State the friction. If the push is then reduced to 20 N, describe what happens. 3

16. Explain why the friction between a book and the table does not depend much on the area of its face for the same weight. 3

17. Why does an object dropped from a height of 1 m take less time to reach the ground than from 2 m? 3

OR

Explain why a cyclist bends forward when riding against the wind. 3

18. How can a magnetised needle be made? Describe the process. 3

OR

Suggest why oil is placed in car engines and what would happen if there was no oil. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Harpreet pulls a box of mass 20 kg across a floor with a horizontal force of 60 N. The force of friction between the box and the floor is 40 N. When he puts rollers under the box, the friction drops to 10 N. 5
(i) Find the net force on the box in the first case.
(ii) Is the box moving or at rest?
(iii) State the direction of friction.
(iv) Find the net force with rollers and explain why rollers help.

20. A man of weight 700 N stands in a lift at rest. Discuss the forces. Compare the forces when the lift moves up with constant speed and explain why the reading of a weighing machine is unchanged. 5

OR

A comb rubbed on dry hair attracts paper, but not when kept in a humid room. Explain with the ideas of charge, conductors and insulators. 5

END OF QUESTION PAPER — Total: 40 Marks